Lesson Notes

MODULE 7 | LESSON 3

CREDIT MODELING IN A DETERIORATING CLIMATE

Reading Time	20 minutes
Prior Knowledge	basic credit concepts
Keywords	Merton credit model, Vasicek credit model, Single-factor model, Transition risk, Physical risk

In the last two lessons, we used Bayesian networks to model a particular type of climate risk, which of course has financial consequences. In this and the next lesson, we temporarily set aside Bayesian networks to look closely at a model for loan portfolios that takes a larger view of the types of climate risks to credit instruments.

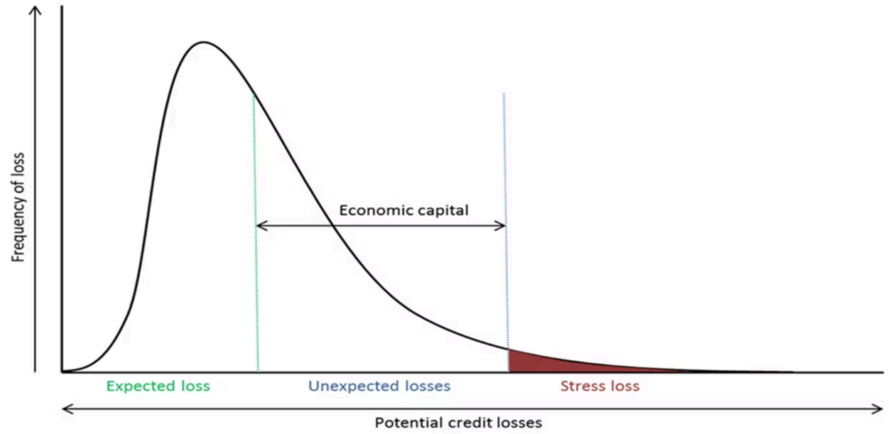
1. Expected (Memory) Loss: A Short Review of Basic Credit Terms

As you will see in this lesson's reading by Garnier et al., the expected loss (EL) of a portfolio of loans is just the sum of the expected losses of the individual loans. The EL for an individual loan is just loan amount (or Exposure at Default, "EAD") multiplied by the Probability of Default (PD) multiplied by the Loss given Default (LGD).

$$EL = EAD * PD * LGD.$$

Just as important as knowing the terms is understanding why this formula actually makes perfect sense, so if it does not yet, convince yourself: If you are owed one million JPY (this is your EAD), but you will only receive 350,000 JPY if the borrower defaults (then your LGD is 35%). The probability that the borrower will default is only 0.5% (this is your PD). The EL is just an expectation: In this case, the expectation is the sum of the two probability-weighted loss amounts: 99.5% probability of zero loss plus a 0.5% chance of 350,000 JPY; that is, 1,750 JPY. All of these terms and their acronyms, as well as how they relate to each other in the formula above, should be familiar to you from the Financial Markets course. If this is not the case, even after this curt review, please take a moment to review as these terms will be used frequently in this and the next lesson as well as the quizzes.

Figure 1: Expected Loss, Unexpected Loss, and Economic Capital



Source: Dolfin, Marina, et al. "Credit Risk Contagion and Systemic Risk on Networks." *MDPI*, 13 Jun 2019, https://www.mdpi.com/2227-7390/7/8/713/htm.

As you know, the mean of the PDF marks the 50% CDF, and you can see in the above figure that indeed half of the probability density lies on one side of the mean and the other half on the other side, so there is a 0.5 probability that the loss will be larger and 0.5 probability that the loss will be smaller than the EL.

2. Expect the Unexpected: Unexpected Loss

As a risk manager, especially at a financial institution, you have to be prepared for losses that greatly exceed such an ordinary loss amount as the EL—not just from a business perspective, but also regulators are likely to demand you reserve much more than this amount.

2.1 The Merton Model

The Merton model, which the Vasicek model builds on, defines a loan default as the event when the borrower's assets at time of the loan maturity are below the borrower's payable contractual obligations.

This is a slight modification of the definition of default as negative equity that was discussed in Financial Markets. Given the fundamental accounting equation:

$$Assets = Liabilities + Equity$$

If equity is negative, then

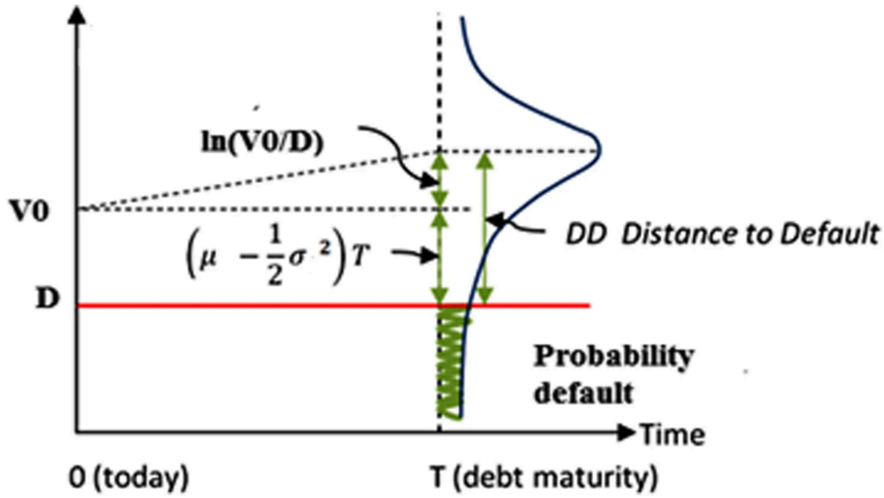
$$Assets - Liabilities = Equity < 0$$

This makes sense since assets are used to pay off debts.

The modification is that, instead of accounting for *all* liabilities, we are only concerned with the currently payable amount or the amount due at the time of the maturity of the loan or loans in question. This also makes sense because not all liabilities are due; longer-term debt can (hopefully) be paid off later.

Of course, all other things being equal, the larger the value of assets relative to the default threshold, the lower the probability of default upon loan maturity. Now assume that the log of assets follows a normal distribution, and we only need to know the standard deviation of the asset value to determine the PD, the probability that the asset value will randomly cross the default threshold at the time of loan maturity. Knowing the mean and the standard deviation, we have specified the (assumed) Normal distribution of the assets and we can straightforwardly calculate the Value-at-Risk (VaR) in the traditional way.

Figure 2: Distance to Default



Source: Dar, Amir, et al. "Estimating Probabilities of Default of Different Firms and the Statistical Tests." *Journal of Global Entrepreneurship Research*, vol. 9, no. 27, 2019, https://link.springer.com/article/10.1186/s40497-019-0152-8.

2.2 The Vasicek Model

The Vasicek model needs the PD as an input, which it can get from the Merton model (or a ratings agency). Knowing the probability of default, it measures the **distance to default** as the number of equivalent standard deviations between zero and the default threshold on the standard Normal pdf. The standard Normal random variable Z represents overall economic conditions in this single-factor model. From here, you will see that traditional Monte Carlo simulation helps determine the 99.9% VaR for a portfolio of loans.

3. Climate Extended Risk Model (CERM)

CERM's extensions to the Vasicek model include the following:

- CERM is a multi-factor model, maintaining the traditional systematic *economic* factor, but also incorporating as systematic risk factors the physical and transition risks associated with the current climate circumstances.
- The LGD is generalized so that it can be calculated stochastically (instead of merely deterministically), to account for how recovery values may change depending on future scenarios. For example, imagine
  - The economy is faltering so potential buyers of assets do not have the income or cash to make capital expenditures such as large asset purchases. In this case, the traditional economic risk factor will likely increase the LGD because defaulted borrowers cannot sell their assets at previously valid prices to help pay down the defaulted loan amount.
  - A fossil-fuel company tries to sell an oil refinery to pay down a defaulted loan in a time that regulators impose steep taxes on the greenhouse gas emissions associated with the refinery process. More than likely, such assets will be worth much less than before such *transition* efforts went into effect. The LGD will have increased due to the lower recovery rate (RR).

    - A manufacturing company needs to sell its processing plant to pay down a defaulted loan, but the location of the plant is either in a region newly prone to flooding or has already suffered major damage due to recent floods. In either case, the physical risks have led to the reduced RR and increased LGD.
- The credit quality migration is also generalized to allow for sensitivity to the same three factors.
    - In a thriving economy, downward credit quality migration may generally be less likely and upward migration more likely.
    - The credit quality of a solar company may not be as likely to fall in a world where demand for solar energy is high, due to its sensitivity to the transition risk factor.
    - The credit quality of a real estate company with large exposures to geographies vulnerable to drought or flooding may be more susceptible to decline as climate catastrophes increase and/or intensify.

4. Conclusion

In this lesson, we reviewed credit fundamentals as well as the more sophisticated Merton and Vasicek models—and their extension in the form of the Climate Extended Risk Model. In the next lesson, we look at how to implement aspects of this model in Python.

References

- Dar, Amir, et al. "Estimating Probabilities of Default of Different Firms and the Statistical Tests." *Journal of Global Entrepreneurship Research*, vol. 9, no. 27, 2019, https://link.springer.com/article/10.1186/s40497-019-0152-8.
- Dolfin, Marina, et al. "Credit Risk Contagion and Systemic Risk on Networks." *MDPI*, 13 Jun 2019, https://www.mdpi.com/2227-7390/7/8/713/htm.
- Garnier, Josselin, et al. "The Climate Extended Risk Model (CERM)." *arXiv*, 10 Apr 2022, https://arxiv.org/abs/2103.03275.